

Naveed Ahmed Khan

Principal Backend Engineer – Distributed Systems

📍 Remote (Global / North America Overlap) | ✉️ naveed.khan@example.com | 🔗 linkedin.com/in/naveed-ahmed-khan | 🌐 github.com/naveed-khan

🎯 EXECUTIVE PROFILE

Principal Backend Engineer with **10 years of experience** designing, scaling, and operating high-throughput **distributed systems, event-driven microservices, and asynchronous messaging architectures**. Deep technical mastery in **C#, .NET 8 / ASP.NET Core, Azure Service Bus, SQL Server, MongoDB, and Kubernetes**. Proven history of optimizing systems processing **10M+ monthly database transactions**, reducing API response latency by **45%**, and slashing query response times by **50%**. Certified Azure Solutions Architect (AZ-305) and Kubernetes Application Developer (CKAD) with a strong foundation in Domain-Driven Design (DDD), Clean Architecture, and high-concurrency systems.

🔧 CORE COMPETENCIES & ARCHITECTURE EXPERTISE

- **Backend & Distributed Systems:** C#, .NET 8 / .NET Core, ASP.NET Core Web API, Node.js, Python, Asynchronous I/O, High-Concurrency Threading, CQRS, Event Sourcing, Domain-Driven Design (DDD), Clean Architecture.
- **Messaging & Event Streaming:** Azure Service Bus (Queues, Topics, Subscriptions), Dead-Letter Routing, Publish/Subscribe Patterns, Event-Driven Architectures, Asynchronous Worker Queues.
- **Databases & High-Scale Storage:** SQL Server (T-SQL, Execution Plans, Index Tuning, Partitioning), PostgreSQL, MongoDB (Aggregation Pipelines, Compound Indexes), Amazon DocumentDB.
- **Cloud & Container Infrastructure:** Microsoft Azure (App Services, Functions, Key Vault, APIM, Blob Storage), AWS (Lambda, EKS, CloudWatch), Docker, Kubernetes (AKS), Helm, CI/CD Automation (GitLab, GitHub Actions).
- **Reliability & Observability:** Application Insights, Distributed Tracing, Structured Logging, Health Checks, APM Profiling, Apache JMeter Performance Testing.
- **Engineering Leadership:** Architectural Decision Records (ADRs), System Design Governance, Technical Mentorship (15+ engineers), Zero-Downtime Migration Strategies (Strangler Fig).

👛 PROFESSIONAL EXPERIENCE

HealthTech & Cloud SaaS Enterprise (*Confidential*)

Technical Lead / Lead Software Engineer | January 2023 – Present

High-Availability Cloud Microservices & Distributed Architecture | Squad: 8–10 Engineers

- **Distributed Microservices Architecture:** Architected and deployed an event-driven microservices platform on **Microsoft Azure** using **C#, .NET 8, and ASP.NET Core Web API**.
- **Asynchronous Event Pipelines:** Implemented high-throughput message routing and pub/sub decoupling using **Azure Service Bus (Queues & Topics)** and **Azure Functions**, eliminating service-to-service contention.
- **System Scale & Latency Optimization:** Scaled backend services to support **15,000+ concurrent enterprise users across 120+ facilities**, reducing overall **API response latency by 45%** through asynchronous processing and Redis caching.
- **Container Orchestration & DevOps:** Led containerization with **Docker** and multi-cluster workload management on **Kubernetes (AKS)**; built zero-downtime CI/CD pipelines reducing deployment windows from **hours to <15 minutes**.
- **Cloud Cost Optimization:** Reduced cloud infrastructure operating expenditure by **30%** through workload auto-scaling and container rightsizing.
- **Architectural Governance:** Authored Architectural Decision Records (ADRs), governed API contract standards, and established automated testing suites achieving **>85% code coverage**.

Global Digital Solutions Consultancy (*Confidential*)

Senior Software Engineer | May 2019 – December 2022

High-Throughput Enterprise Backend Services | US & UK Enterprise Clients

- **High-Throughput API Processing:** Built and maintained distributed RESTful APIs in **C# / .NET Core**, processing **10M+ monthly business transactions** with sub-100ms response times.
- **Database Scalability & Query Tuning:** Optimized complex SQL Server stored procedures, indexing schemes, and execution plans, reducing critical reporting endpoint latency by **40%**.
- **Service Decoupling & Modernization:** Modularized legacy monolithic components into independent, domain-driven REST services.
- **System Quality & Reliability:** Reduced production incidents by **35%** through comprehensive automated test suites and standardized peer review processes.

Enterprise Software Services Provider (*Confidential*)

Software Developer | March 2015 – April 2019

High-Volume Messaging & Telecommunications

- **Messaging Management Platform:** Engineered full-stack messaging and communication engines processing **100,000+ monthly outbound and inbound SMS transactions**.
- **NoSQL Performance Tuning:** Optimized MongoDB schema models and compound indexes, lowering search and pagination response times to **<80ms** (50% speedup).
- **Service Uptime:** Achieved **99.8% API uptime** across legacy and modern services through resilient error handling, retry policies, and monitoring.

FEATURED DISTRIBUTED SYSTEMS PROJECTS

Nursing Home Management SaaS Microservices Platform

- **Domain:** Healthcare Enterprise Operations
- **Tech Stack:** C#, .NET 8, ASP.NET Core, Azure Service Bus, Azure Functions, Kubernetes, Docker, SQL Server.
- **Deliverables:** Re-engineered legacy desktop system into a high-scale event-driven cloud SaaS. Decoupled scheduling, compliance, and timecard domains into independent microservices. Scaled to 15,000+ users with 45% latency improvement and 99.99% availability.

High-Throughput Messaging & SMS Search Engine

- **Domain:** Telecommunications & Real-Time Ingestion
 - **Tech Stack:** Node.js, Express, MongoDB, C#, SQL Server, Redis.
 - **Deliverables:** Engineered an asynchronous message ingestion pipeline capable of indexing 100k+ monthly SMS records. Designed compound indexing strategies to support sub-80ms pagination and multi-parameter filtering across millions of historic records.
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PROFESSIONAL CERTIFICATIONS

- **Microsoft Certified: Azure Solutions Architect Expert (AZ-305)** — *Microsoft (2023)*
 - **Certified Kubernetes Application Developer (CKAD)** — *Cloud Native Computing Foundation (2022)*
 - **Microsoft Certified: Azure Developer Associate (AZ-204)** — *Microsoft (2021)*
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EDUCATION & ACADEMIC HONORS

Bachelor of Engineering (B.E.) — Computer & Information Systems Engineering

NED University of Engineering and Technology, Karachi, Pakistan | 2012 – 2016

- **Academic Standing:** CGPA **3.86 / 4.00** (*High Distinction — Top 5% of Class*)
- **Relevant Coursework:** Distributed Computing & Systems, Database Management Systems (DBMS), Computer Architecture & Operating Systems, Computer Networks.
- **Continuous Learning:** Microservices Architecture & Distributed Systems — *O'Reilly / Coursera (2022)*.